

Test ID	Feature / Operation	Category	Priority	Description	Stimulus Summary	Expected Result	Coverage Bin	Sequence / File	Status	Notes
TC_OR_001	6.1.1 OR	Directed-Valid	High	Confirms plain bitwise OR result.	ap.lor=1; ap.zbb=0; a_in=0xF0F0F0F0; b_in=0x0F0F0F0F	result=0xFFFFFFFF; error=0	OR x zbb=0	tb/sequences/logic_ops/or_valid_seq.sv	Not Started	
TC_OR_002	6.1.1 OR	Directed-Valid	High	Confirms OR-with-inverted-b_in variant (ap.zbb=1).	ap.lor=1; ap.zbb=1; a_in=0xAAAAAAAA; b_in=0x55555555	result=0xFFFFFFFF; error=0	OR x zbb=1	tb/sequences/logic_ops/or_invert_seq.sv	Not Started	
TC_OR_003	6.1.1 OR	Directed-Error	High	Guard violation: a second ap.* field active alongside ap.lor.	ap.lor=1; ap.lxor=1 (conflicting field); csr_ren_in=0	result=0; error=1	OR x guard-violation	tb/sequences/error_injection/or_conflict_seq.sv	Not Started	
TC_XOR_001	6.1.2 XOR	Directed-Valid	High	Confirms plain bitwise XOR result.	ap.lxor=1; ap.zbb=0; a_in=0xF0F0F0F0; b_in=0x0F0F0F0F	result=0xFFFFFFFF; error=0	XOR x zbb=0	tb/sequences/logic_ops/xor_valid_seq.sv	Not Started	
TC_XOR_002	6.1.2 XOR	Directed-Valid	High	Confirms XOR-with-inverted-b_in variant (ap.zbb=1).	ap.lxor=1; ap.zbb=1; a_in=0xAAAAAAAA; b_in=0x55555555	result=0x00000000; error=0	XOR x zbb=1	tb/sequences/logic_ops/xor_invert_seq.sv	Not Started	
TC_XOR_003	6.1.2 XOR	Directed-Error	High	Guard violation: csr_ren_in asserted alongside ap.lxor.	ap.lxor=1; csr_ren_in=1	result=0; error=1	XOR x guard-violation	tb/sequences/error_injection/xor_csr_conflict_seq.sv	Not Started	
TC_SRL_001	6.2.1 SRL	Directed-Valid	High	Nominal logical right shift.	ap.srl=1; a_in=0xF0000000; b_in[4:0]=4	result=0xF0000000; error=0	SRL x mid-range shift	tb/sequences/shift_ops/srl_valid_seq.sv	Not Started	
TC_SRL_002	6.2.1 SRL	Directed-Corner	Medium	Boundary shift amounts: 0 (no-op) and 31 (max).	ap.srl=1; b_in[4:0]=0 and b_in[4:0]=31, a_in=0xFFFFFFFF	shift=0: result=a_in unchanged; shift=31: result=0x00000001; error=0	SRL x {0,31}	tb/sequences/shift_ops/srl_corner_seq.sv	Not Started	
TC_SRL_003	6.2.1 SRL	Directed-Error	Medium	Guard violation on SRL.	ap.srl=1; ap.sra=1 (conflicting field)	result=0; error=1	SRL x guard-violation	tb/sequences/error_injection/srl_conflict_seq.sv	Not Started	
TC_SRA_001	6.2.2 SRA	Directed-Valid	High	Nominal arithmetic right shift, negative operand.	ap.sra=1; a_in=0xF0000000; b_in[4:0]=4	result=0xFF000000 (sign-extended); error=0	SRA x negative operand	tb/sequences/shift_ops/sra_valid_seq.sv	Not Started	
TC_SRA_002	6.2.2 SRA	Directed-Corner	Medium	Max shift amount fully sign-fills the result.	ap.sra=1; a_in=0x80000000; b_in[4:0]=31	result=0xFFFFFFFF; error=0	SRA x shift=31	tb/sequences/shift_ops/sra_corner_seq.sv	Not Started	
TC_SRA_003	6.2.2 SRA	Directed-Error	Medium	Guard violation on SRA.	ap.sra=1; ap.ror=1 (conflicting field)	result=0; error=1	SRA x guard-violation	tb/sequences/error_injection/sra_conflict_seq.sv	Not Started	
TC_ROR_001	6.2.3 ROR	Directed-Valid	High	Nominal rotate-right result.	ap.ror=1; a_in=0x89ABCDEF; b_in[4:0]=4	result=0xF89ABCDE; error=0	ROR x mid-range amount	tb/sequences/shift_ops/ror_valid_seq.sv	Not Started	
TC_ROR_002	6.2.3 ROR	Directed-Corner	Medium	Boundary amounts: 0 (no-op) and 31 (near-full rotation).	ap.ror=1; b_in[4:0]=0 and 31, a_in=0x00000001	shift=0: result unchanged; shift=31: result=0x00000002; error=0	ROR x {0,31}	tb/sequences/shift_ops/ror_corner_seq.sv	Not Started	
TC_ROR_003	6.2.3 ROR	Directed-Error	Medium	Guard violation on ROR.	ap.ror=1; ap.binv=1 (conflicting field)	result=0; error=1	ROR x guard-violation	tb/sequences/error_injection/ror_conflict_seq.sv	Not Started	
TC_BINV_001	6.2.4 BINV	Directed-Valid	High	Nominal single-bit invert, mid-range position.	ap.binv=1; a_in=0xFFFFFFFF; b_in[4:0]=2	result=0xFFFFF8; error=0	BINV x mid-range bit	tb/sequences/bit_ops/binv_valid_seq.sv	Not Started	
TC_BINV_002	6.2.4 BINV	Directed-Corner	Medium	Boundary bit positions 0 and 31.	ap.binv=1; b_in[4:0]=0 and 31, a_in=0x00000000	bit0: result=0x00000001; bit31: result=0x80000000; error=0	BINV x {bit0,bit31}	tb/sequences/bit_ops/binv_corner_seq.sv	Not Started	
TC_BINV_003	6.2.4 BINV	Directed-Error	Medium	Guard violation on BINV.	ap.binv=1; csr_ren_in=1	result=0; error=1	BINV x guard-violation	tb/sequences/error_injection/binv_csr_conflict_seq.sv	Not Started	
TC_SH2ADD_001	6.3.1 SH2ADD	Directed-Valid	High	Nominal shift-left-2-and-add, Zba mode correctly asserted.	ap.sh2add=1; ap.zba=1; a_in=4; b_in=7	result=23; error=0	SH2ADD x valid	tb/sequences/zba_ops/sh2add_valid_seq.sv	Not Started	
TC_SH2ADD_002	6.3.1 SH2ADD	Directed-Error	High	Op-defining guard: ap.zba not asserted.	ap.sh2add=1; ap.zba=0	result=0; error=1	SH2ADD x zba=0	tb/sequences/zba_ops/sh2add_no_zba_seq.sv	Not Started	
TC_SH2ADD_003	6.3.1 SH2ADD	Directed-Error	Medium	Generic guard violation: conflicting field + CSR read.	ap.sh2add=1; ap.zba=1; ap.lor=1; csr_ren_in=1	result=0; error=1	SH2ADD x guard-violation	tb/sequences/error_injection/sh2add_conflict_seq.sv	Not Started	
TC_SUB_001	6.4.1 SUB	Directed-Valid	High	Nominal subtraction.	ap.sub=1; a_in=20; b_in=7	result=13; error=0	SUB x nominal	tb/sequences/base/sub_valid_seq.sv	Not Started	
TC_SUB_002	6.4.1 SUB	Directed-Corner	Medium	Underflow wraps per two's-complement (0 - 1).	ap.sub=1; a_in=0; b_in=1	result=0xFFFFFFFF; error=0	SUB x underflow	tb/sequences/base/sub_corner_seq.sv	Not Started	
TC_SUB_003	6.4.1 SUB	Directed-Error	High	Op-defining guard: plain SUB with ap.zba mistakenly asserted.	ap.sub=1; ap.zba=1	result=0; error=1	SUB x zba-misuse	tb/sequences/error_injection/sub_zba_misuse_seq.sv	Not Started	
TC_SLT_001	6.4.2 SLT/SLTU	Directed-Valid	High	Signed SLT, negative < positive.	ap.slt=1; ap.sub=1; ap.unsign=0; a_in=-2; b_in=1	result=1; error=0	SLT x signed	tb/sequences/base/slt_signed_seq.sv	Not Started	
TC_SLT_002	6.4.2 SLT/SLTU	Directed-Valid	High	Unsigned SLTU — same bit pattern as TC_SLT_001 compares as large positive, not Equal operands must yield 0 in both modes.	ap.slt=1; ap.sub=1; ap.unsign=1; a_in=0xFFFFFFFF; b_in=1	result=0; error=0	SLT x unsigned	tb/sequences/base/slt_unsigned_seq.sv	Not Started	
TC_SLT_003	6.4.2 SLT/SLTU	Directed-Corner	Medium	Equal operands must yield 0 in both modes.	ap.slt=1; ap.sub=1; a_in=5; b_in=5; ap.unsign=0 then 1	result=0 in both modes; error=0	SLT x equal-operands	tb/sequences/base/slt_equal_seq.sv	Not Started	
TC_SLT_004	6.4.2 SLT/SLTU	Directed-Error	Medium	Guard violation on SLT.	ap.slt=1; ap.sub=1; ap.lor=1 (conflicting field)	result=0; error=1	SLT x guard-violation	tb/sequences/error_injection/slt_conflict_seq.sv	Not Started	
TC_CTZ_001	6.5.1 CTZ	Directed-Valid	High	Nominal trailing-zero count.	ap.ctz=1; a_in=0x0000100	result=8; error=0	CTZ x mid-range	tb/sequences/count_ops/ctz_valid_seq.sv	Not Started	
TC_CTZ_002	6.5.1 CTZ	Directed-Corner	High	All-zero input — the corrected value (was wrongly 0 in spec v1.1). High priority: prior spec had a must-be-zero guard.	ap.ctz=1; a_in=0x00000000	result=32; error=0	CTZ x all-zero	tb/sequences/count_ops/ctz_zero_seq.sv	Not Started	Verifies Specification v1.2 correction over v1.1
TC_CTZ_003	6.5.1 CTZ	Directed-Corner	Medium	All-ones input — bit 0 already set, count is 0.	ap.ctz=1; a_in=0xFFFFFFFF	result=0; error=0	CTZ x all-ones	tb/sequences/count_ops/ctz_ones_seq.sv	Not Started	
TC_CTZ_004	6.5.1 CTZ	Directed-Error	Medium	Guard violation on CTZ.	ap.ctz=1; csr_ren_in=1	result=0; error=1	CTZ x guard-violation	tb/sequences/error_injection/ctz_csr_conflict_seq.sv	Not Started	
TC_CPOP_001	6.5.2 CPOP	Directed-Valid	High	Nominal population count, mixed bits.	ap.cpop=1; a_in=0xF0F0F0F0	result=16; error=0	CPOP x mixed	tb/sequences/count_ops/cpop_valid_seq.sv	Not Started	
TC_CPOP_002	6.5.2 CPOP	Directed-Corner	Medium	Boundary counts: all-zero and all-one inputs.	ap.cpop=1; a_in=0x00000000 and 0xFFFFFFFF	result=0 and 32 respectively; error=0	CPOP x {0,32}	tb/sequences/count_ops/cpop_corner_seq.sv	Not Started	
TC_CPOP_003	6.5.2 CPOP	Directed-Error	Medium	Guard violation on CPOP.	ap.cpop=1; ap.pack=1 (conflicting field)	result=0; error=1	CPOP x guard-violation	tb/sequences/error_injection/cpop_conflict_seq.sv	Not Started	
TC_SEXTB_001	6.6.1 SEXT.B	Directed-Valid	Medium	Positive byte value (bit 7 = 0) — upper bits stay 0.	ap.sixt_b=1; a_in=0x0000007F	result=0x0000007F; error=0	SEXT.B x positive	tb/sequences/sign_extend_ops/sextb_pos_seq.sv	Not Started	
TC_SEXTB_002	6.6.1 SEXT.B	Directed-Corner	High	Negative byte value (bit 7 = 1) — sign-bit replication is the core function being tested.	ap.sixt_b=1; a_in=0x000000FF	result=0xFFFFFFFF; error=0	SEXT.B x negative	tb/sequences/sign_extend_ops/sextb_neg_seq.sv	Not Started	
TC_SEXTB_003	6.6.1 SEXT.B	Directed-Error	Low	Guard violation on SEXT.B.	ap.sixt_b=1; csr_ren_in=1	result=0; error=1	SEXT.B x guard-violation	tb/sequences/error_injection/sextb_csr_conflict_seq.sv	Not Started	

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TC_MAX_001	6.7.1 MAX	Directed-Valid	High	Nominal max, b_in larger.	ap.max=1; ap.sub=1; a_in=10; b_in=20	result=20; error=0	MAX x b-larger	tb/sequences/minmax_ops/max_valid_seq.sv	Not Started	
TC_MAX_002	6.7.1 MAX	Directed-Valid	Medium	Operand order reversed — confirms comparator, not a fixed operand preference.	ap.max=1; ap.sub=1; a_in=20; b_in=10	result=20; error=0	MAX x a-larger	tb/sequences/minmax_ops/max_reversed_seq.sv	Not Started	
TC_MAX_003	6.7.1 MAX	Directed-Corner	Medium	Negative vs. positive — confirms signed comparison, not raw bit magnitude.	ap.max=1; ap.sub=1; a_in=-1; b_in=1	result=1; error=0	MAX x signed-boundary	tb/sequences/minmax_ops/max_signed_seq.sv	Not Started	
TC_MAX_004	6.7.1 MAX	Directed-Error	Medium	Guard violation on MAX.	ap.max=1; ap.sub=1; ap.lxor=1 (conflicting field)	result=0; error=1	MAX x guard-violation	tb/sequences/error_injection/max_conflict_seq.sv	Not Started	
TC_PACK_001	6.8.1 PACK	Directed-Valid	Medium	Nominal pack of two half-words.	ap.pack=1; a_in=0x00001234; b_in=0x00005678	result=0x56781234; error=0	PACK x nominal	tb/sequences/pack_ops/pack_valid_seq.sv	Not Started	
TC_PACK_002	6.8.1 PACK	Directed-Corner	Low	Boundary half-word patterns (all-0 / all-1).	ap.pack=1; a_in=0x0000FFFF; b_in=0x00000000	result=0x0000FFFF; error=0	PACK x boundary	tb/sequences/pack_ops/pack_corner_seq.sv	Not Started	
TC_PACK_003	6.8.1 PACK	Directed-Error	Low	Guard violation on PACK.	ap.pack=1; csr_ren_in=1	result=0; error=1	PACK x guard-violation	tb/sequences/error_injection/pack_csr_conflict_seq.sv	Not Started	
TC_GREV_001	6.8.2 GREV	Directed-Valid	High	Nominal byte-reverse, the one implemented encoding.	ap.grev=1; a_in=0x12345678; b_in[4:0]=24	result=0x78563412; error=0	GREV x b_in=24	tb/sequences/pack_ops/grev_valid_seq.sv	Not Started	
TC_GREV_002	6.8.2 GREV	Directed-Error	High	ASSUMPTION (Spec §7.1, not design-team-confirmed); undefined encoding treated as	ap.grev=1; b_in[4:0]=5 (≠ 24)	result=0; error=1 — per adopted assumption, re-verify if design team answers differently	GREV x undefined-encoding	tb/sequences/pack_ops/grev_undefined_seq.sv	Not Started	See CLARIF-004
TC_GREV_003	6.8.2 GREV	Directed-Error	Medium	Generic guard violation on GREV.	ap.grev=1; b_in[4:0]=24; csr_ren_in=1	result=0; error=1	GREV x guard-violation	tb/sequences/error_injection/grev_csr_conflict_seq.sv	Not Started	
TC_CSR_001	6.9.1 CSR bypass read	CSR	High	Pure CSR bypass read — directly validates the CSR-conflict scoping assumption (Spec §7.2).	csr_ren_in=1; ap=0 (all fields clear); csr_rddata_in=0xABCD1234	result=0xABCD1234; error=0	CSR x bypass-read	tb/sequences/csr_ops/csr_bypass_read_seq.sv	Not Started	See CLARIF-005
TC_CSR_002	6.9.2 CSR write	CSR	Medium	CSR write, immediate form.	ap.csr_write=1; ap.csr_imm=1; b_in=0x11112222	result=0x11112222; error=0	CSR x write-imm	tb/sequences/csr_ops/csr_write_imm_seq.sv	Not Started	
TC_CSR_003	6.9.2 CSR write	CSR	Medium	CSR write, register form.	ap.csr_write=1; ap.csr_imm=0; a_in=0x33334444	result=0x33334444; error=0	CSR x write-reg	tb/sequences/csr_ops/csr_write_reg_seq.sv	Not Started	
TC_CSR_004	§5 Chip-wide error	CSR	High	Chip-wide conflict: CSR read requested simultaneously with a bit-manip op	csr_ren_in=1; ap.lor=1	result=0; error=1	Chip-wide x CSR-conflict	tb/sequences/csr_ops/csr_bitmanip_conflict_seq.sv	Not Started	
TC_TIME_001	§3.1 Latency	Timing	High	Confirms 1-cycle latency: result_ff updates the cycle after valid_in, not the same cycle.	valid_in=1 at cycle N with a valid OR request; sample result_ff at N and N+1	result_ff unchanged at N; updated result visible at N+1; error=0	Timing x latency	tb/sequences/reset_ops/latency_check_seq.sv	Not Started	
TC_TIME_002	§3.2 valid_in gating	Timing	High	Confirms result_ff holds when valid_in=0, while error stays live.	Drive a valid op with valid_in=1, then valid_in=0 with changing/garbage ap fields	result_ff holds prior value; error reflects the live (garbage) inputs each cycle	Timing x valid_in=0-hold	tb/sequences/reset_ops/valid_in_hold_seq.sv	Not Started	
TC_TIME_003	§3.2 Back-to-back valid	Timing	Medium	Confirms correct pipelining with valid_in=1 every cycle, different op each cycle.	valid_in=1 for 3 consecutive cycles, OR then XOR then SRL	result_ff shows each op's result exactly one cycle after its inputs; error=0 throughout	Timing x back-to-back	tb/sequences/reset_ops/back_to_back_seq.sv	Not Started	
TC_RST_001	§3.3 Reset	Reset	High	Confirms synchronous reset clears both outputs.	Drive a valid non-zero result, then assert rst_l=0 for one cycle	result_ff=0; error=0 immediately after reset	Reset x clear	tb/sequences/reset_ops/reset_clear_seq.sv	Not Started	
TC_RST_002	§3.3 Reset	Reset	Medium	Reset asserted while valid_in=1 — reset must win.	rst_l=0 and valid_in=1 simultaneously with a valid op driven	result_ff=0; error=0 (reset overrides the concurrent valid operation)	Reset x mid-operation	tb/sequences/reset_ops/reset_mid_op_seq.sv	Not Started	
TC_SCAN_001	§3.4 scan_mode	Reset	Low	Confirms scan_mode has no functional effect when toggled during normal operation.	Toggle scan_mode 0->1->0 while driving a repeated valid OR operation	result_ff and error unaffected by scan_mode transitions	Reset x scan_mode-inert	tb/sequences/reset_ops/scan_mode_seq.sv	Not Started	
TC_RAND_001	All operations	Random	High	Constrained-random: legal single-operation encodings across the full operand range, for non-degenerated operands.	Randomize a_in, b_in fully; randomize exactly one legal ap.* field per transaction; N=[TBD] transactions	Reference-model match on result_ff and error every transaction	Cross: operation x operand-range	tb/sequences/base/random_legal_seq.sv	Not Started	
TC_RAND_002	All operations	Random	Medium	Constrained-random weighted toward corner values (0, all-1s, max shift/bit-index amounts).	Weighted randomization favoring 0x00000000, 0xFFFFFFFF, and shift/index boundary values	Reference-model match every transaction; closes remaining corner-case coverage bins	Cross: operation x corner-weighted	tb/sequences/base/random_corner_weighted_seq.sv	Not Started	
TC_RAND_003	All operations	Random	Medium	Constrained-random illegal/conflicting ap field combinations — error-path stress.	Randomize 2+ simultaneous ap.* fields and/or csr_ren_in with a bit-manip field	result_ff=0 and error=1 on every illegal combination generated	Cross: guard-violation x random-combinations	tb/sequences/error_injection/random_error_seq.sv	Not Started	
TC_TBD_001	§8 ap.min/ap.clz/ap.rol	Scope-TBD	Low	BLOCKED: fields are functionally active in RTL but have no behavior specification. No generated values can be written until the design is complete.	N/A — blocked	BLOCKED — pending Specification §8 / clarification log resolution	N/A	N/A	Not Started	See CLARIF, Verification Plan Risk R-3
TC_TBD_002	§8 ap.bset/ap.bclr	Scope-TBD	Low	BLOCKED: same reason as TC_TBD_001.	N/A — blocked	BLOCKED — pending clarification	N/A	N/A	Not Started	See Verification Plan Risk R-3
TC_TBD_003	§8 ap.packu/ap.packh/ap.gorc	Scope-TBD	Low	BLOCKED: same reason as TC_TBD_001.	N/A — blocked	BLOCKED — pending clarification	N/A	N/A	Not Started	See Verification Plan Risk R-3